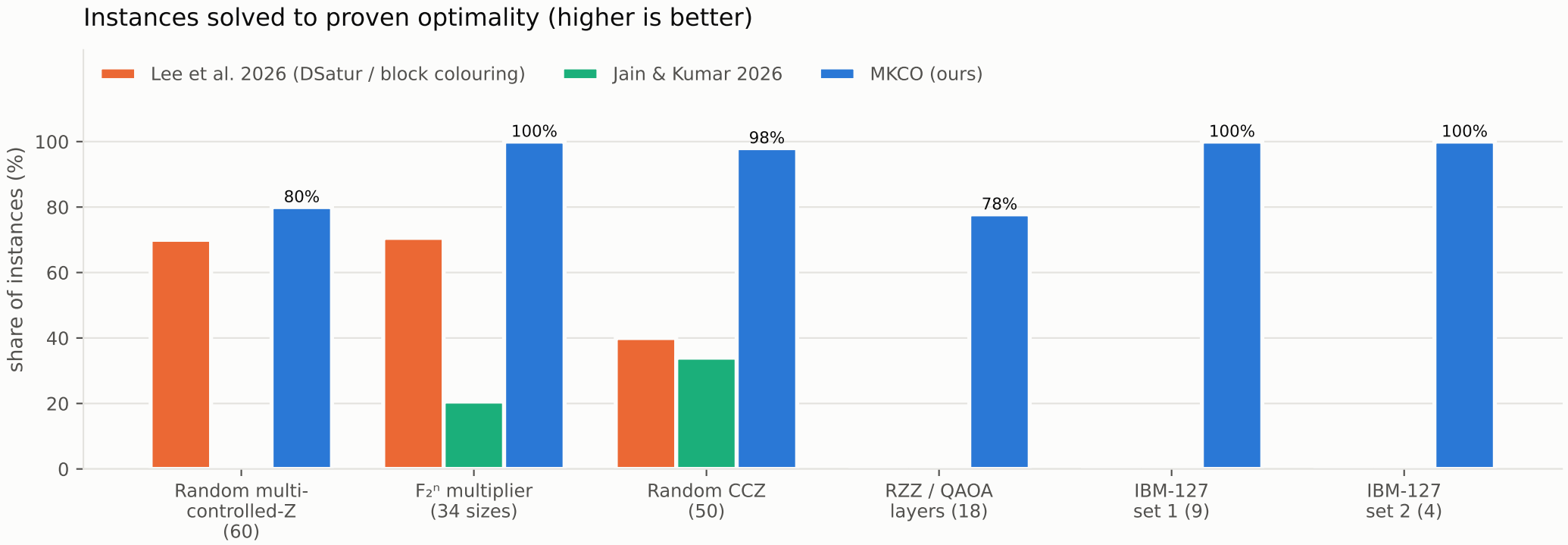
Instances solved to proven optimality (higher is better)
Lee et al. 2026 (DSatur / block colouring)
Jain & Kumar 2026
MKCO (ours)
share of instances (%)
0
20
40
60
80
100
80%
100%
98%
78%
100%
100%
Random multi-controlled-Z (60)
F₂ⁿ multiplier (34 sizes)
Random CCZ (50)
RZZ / QAOA layers (18)
IBM-127 set 1 (9)
IBM-127 set 2 (4)
IBM sets: only MKCO issues a proof (Lee and JK reach no / equal depth without a certificate). RZZ: reaching the degree bound Δ.